complete java class code example of each combination of intermediate and terminal operations

filter() boolean checks + anymatch() existence checks

filter() null checks + anymatch() existence checks

filter() comparisons + anymatch() existence checks

filter() method calls returning boolean + anymatch() existence checks

distinct() rely on equals() / hashcode() + anymatch() existence checks

distinct() immutable objects + anymatch() existence checks

map() transform objects + anymatch() existence checks

map() create new objects + anymatch() existence checks

map() call pure functions + anymatch() existence checks

flatmap() return stream + anymatch() existence checks

flatmap() split data + anymatch() existence checks

flatmap() expand collections + anymatch() existence checks

sorted() natural ordering + anymatch() existence checks

sorted() custom comparator + anymatch() existence checks

sorted() multi-field compare + anymatch() existence checks

peek() logging + anymatch() existence checks

peek() debugging + anymatch() existence checks

peek() metrics + anymatch() existence checks

Below are **all 18 programs converted to Java 1.5 compatible code**.

Since Java 1.5:

- ❌ No Stream API
- ❌ No Optional
- ❌ No lambda expressions
- ❌ No method references
- ❌ No List.of(), Map.of()
- ❌ No Objects.hash()
- ❌ No Comparator.comparing()

We replace everything with:

- ✔ for-each loops
- ✔ Iterator
- ✔ Collections / Arrays
- ✔ HashSet for distinct
- ✔ Manual sorting with Comparator
- ✔ Null checks
- ✔ Primitive counters instead of AtomicInteger

Below are **complete, standalone Java class examples** for **each requested Stream combination**, where:

- **Terminal operation = anyMatch()**
- Purpose = **existence checks** (returns true if **any element** satisfies the predicate)
- Demonstrates **short-circuiting** (stops on first match)
- Java **8+** compatible
- No collectors / reduce / custom aggregation

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1 filter() boolean checks + anyMatch() — Existence check

```
import java.util.*;

public class FilterBooleanAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(1, 3, 4, 7).stream()
            .filter(n -> n > 0)
            .anyMatch(n -> n % 2 == 0);

      System.out.println(exists);
   }
}
```

---

2 filter() null checks + anyMatch() — Existence check

```
import java.util.*;

public class FilterNullAnyMatch {
   public static void main(String[] args) {
      boolean exists = Arrays.asList(null, "Anurag", "Amit").stream()
            .filter(Objects::nonNull)
            .anyMatch(s -> s.startsWith("A"));

      System.out.println(exists);
   }
}
```

---

3 filter() comparisons + anyMatch() — Existence check

```
import java.util.*;

public class FilterComparisonAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(20000, 45000, 60000).stream()
            .filter(s -> s >= 30000)
            .anyMatch(s -> s > 50000);

      System.out.println(exists);
   }
}
```

---

4 filter() method calls returning boolean + anyMatch() — Existence check

```
import java.util.*;

public class FilterMethodAnyMatch {

   static class User {
      boolean active;
      User(boolean active) { this.active = active; }
      boolean isActive() { return active; }
   }

   public static void main(String[] args) {
      boolean exists = List.of(
            new User(false),
            new User(true)
      ).stream()
       .filter(User::isActive)
       .anyMatch(User::isActive);

      System.out.println(exists);
   }
}
```

---

5 distinct() rely on equals() / hashCode() + anyMatch() — Existence check

```
import java.util.*;

public class DistinctEqualsAnyMatch {

   static class Person {
      String name;
      Person(String name) { this.name = name; }

      @Override public boolean equals(Object o) {
         return o instanceof Person p && name.equals(p.name);
      }
      @Override public int hashCode() {
         return Objects.hash(name);
      }
   }

   public static void main(String[] args) {
      boolean exists = List.of(
            new Person("Anurag"),
            new Person("Anurag"),
            new Person("Ravi")
      ).stream()
       .distinct()
       .anyMatch(p -> p.name.equals("Ravi"));

      System.out.println(exists);
   }
}
```

---

6 distinct() immutable objects + anyMatch() — Existence check

```
import java.util.*;

public class DistinctImmutableAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of("Java", "Java", "Python").stream()
            .distinct()
            .anyMatch(lang -> lang.startsWith("P"));

      System.out.println(exists);
   }
}
```

---

7 map() transform objects + anyMatch() — Existence check

```
import java.util.*;

public class MapTransformAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of("anurag", "amit").stream()
            .map(String::toUpperCase)
            .anyMatch(s -> s.equals("AMIT"));

      System.out.println(exists);
   }
}
```

---

8 map() create new objects + anyMatch() — Existence check

```
import java.util.*;

public class MapCreateObjectAnyMatch {

   static class UserDTO {
      String name;
      UserDTO(String name) { this.name = name; }
   }

   public static void main(String[] args) {
      boolean exists = List.of("Anurag", "Amit").stream()
            .map(UserDTO::new)
            .anyMatch(dto -> dto.name.equals("Amit"));

      System.out.println(exists);
   }
}
```

---

9 map() call pure functions + anyMatch() — Existence check

```
import java.util.*;

public class MapPureFunctionAnyMatch {

   static int square(int n) { return n * n; }

   public static void main(String[] args) {
      boolean exists = List.of(2, 3, 4).stream()
            .map(MapPureFunctionAnyMatch::square)
            .anyMatch(n -> n == 16);

      System.out.println(exists);
   }
}
```

---

10 flatMap() return Stream + anyMatch() — Existence check

```
import java.util.*;

public class FlatMapStreamAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(
            List.of(1, 2),
            List.of(3, 4)
      ).stream()
       .flatMap(List::stream)
       .anyMatch(n -> n == 3);

      System.out.println(exists);
   }
}
```

---

11 flatMap() split data + anyMatch() — Existence check

```
import java.util.*;

public class FlatMapSplitAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of("Java Stream", "Flat Map").stream()
            .flatMap(s -> Arrays.stream(s.split(" ")))
            .anyMatch(word -> word.equals("Stream"));

      System.out.println(exists);
   }
}
```

---

12 flatMap() expand collections + anyMatch() — Existence check

```
import java.util.*;

public class FlatMapExpandAnyMatch {
   public static void main(String[] args) {
      boolean exists = Map.of(
            "A", List.of("1", "2"),
            "B", List.of("3")
      ).values().stream()
       .flatMap(Collection::stream)
       .anyMatch(v -> v.equals("2"));

      System.out.println(exists);
   }
}
```

---

13 sorted() natural ordering + anyMatch() — Existence check

```
import java.util.*;

public class SortedNaturalAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(5, 1, 3).stream()
            .sorted()
            .anyMatch(n -> n == 1);

      System.out.println(exists);
   }
}
```

---

14 sorted() custom comparator + anyMatch() — Existence check

```
import java.util.*;

public class SortedCustomAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of("Anurag", "Ravi", "Amit").stream()
            .sorted(Comparator.reverseOrder())
            .anyMatch(name -> name.equals("Amit"));

      System.out.println(exists);
   }
}
```

---

15 sorted() multi-field compare + anyMatch() — Existence check

```
import java.util.*;

public class SortedMultiFieldAnyMatch {

   static class Employee {
      String name;
      int age;
      Employee(String name, int age) {
         this.name = name; this.age = age;
      }
   }

   public static void main(String[] args) {
      boolean exists = List.of(
            new Employee("Anurag", 30),
            new Employee("Anurag", 25),
            new Employee("Ravi", 35)
      ).stream()
       .sorted(Comparator
             .comparing((Employee e) -> e.name)
             .thenComparingInt(e -> e.age))
       .anyMatch(e -> e.age == 25);

      System.out.println(exists);
   }
}
```

---

16 peek() logging + anyMatch() — Existence check

```
import java.util.*;

public class PeekLoggingAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(1, 2, 3).stream()
            .peek(n -> System.out.println("Checking: " + n))
            .anyMatch(n -> n == 2);

      System.out.println(exists);
   }
}
```

---

17 peek() debugging + anyMatch() — Existence check

```
import java.util.*;

public class PeekDebugAnyMatch {
   public static void main(String[] args) {
      boolean exists = List.of(10, 20, 30).stream()
            .peek(n -> {
               if (n == 20) System.out.println("Debug breakpoint");
            })
            .anyMatch(n -> n == 20);
      System.out.println(exists);
   }
}
```

---

18 peek() metrics + anyMatch() — Existence check

```
import java.util.*;
import java.util.concurrent.atomic.AtomicInteger;

public class PeekMetricsAnyMatch {
   public static void main(String[] args) {
      AtomicInteger processed = new AtomicInteger();

      boolean exists = List.of(5, 10, 15).stream()
            .peek(n -> processed.incrementAndGet())
            .anyMatch(n -> n == 10);

      System.out.println("Exists: " + exists);
      System.out.println("Processed elements: " + processed.get());
   }
}
```